

MILFORD HAVEN PORT AUTHORITY, United Kingdom

WMO: 036040

Lat: 51.700N

Lon: 5.050W

Elev: 105

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.0	32.5	22.7	17.1	33.0	25.5	19.6	35.0	38.0	46.1	34.2	46.6	8.4	50	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.8	72.6	63.5	69.4	62.0	67.2	60.9	64.9	69.9	63.5	67.3	62.3	65.4	8.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	86.7	65.8	62.0	83.5	64.6	61.0	80.5	63.7	29.9	70.1	28.9	67.5	28.1	65.5	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.3	27.6	24.6	DB	27.3	77.9	2.8	3.3	25.3	80.2	23.7	82.2	22.1	84.0	20.1	86.4	(4)
(5)			WB	25.5	68.0	2.8	2.3	23.5	69.7	21.9	71.0	20.3	72.3	18.3	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	43.8	43.7	45.3	48.6	53.3	57.6	61.0	61.1	58.2	54.0	48.9	45.2	(6)
(7)		DBStd	7.54	4.98	4.31	3.98	3.92	3.98	3.57	3.37	2.85	2.98	3.88	4.54	5.38	(7)
(8)		HDD50	844	193	178	149	68	12	0	0	0	0	12	72	160	(8)
(9)		HDD65	4849	657	597	610	492	364	224	133	127	205	341	484	615	(9)
(10)		CDD50	1487	2	1	3	26	113	229	341	343	245	136	38	10	(10)
(11)		CDD65	17	0	0	0	0	0	3	9	5	0	0	0	0	(11)
(12)		CDH74	60	0	0	0	0	0	10	31	19	0	0	0	0	(12)
(13)		CDH80	3	0	0	0	0	0	0	1	2	0	0	0	0	(13)
(14)	Wind	WSAvg	12.0	14.1	13.6	12.2	11.2	10.8	10.1	9.7	10.3	11.1	13.3	13.4	14.1	(14)
(15)	Precipitation	PrecAvg	33.9	3.5	2.6	2.6	2.1	1.9	2.0	1.9	2.9	2.9	3.7	4.0	3.8	(15)
(16)		PrecMax	44.2	7.2	6.1	6.2	5.4	4.7	4.5	5.4	7.1	5.8	8.2	10.1	7.0	(16)
(17)		PrecMin	24.3	0.5	0.1	0.4	0.4	0.2	0.2	0.1	0.0	0.4	0.7	1.2	1.1	(17)
(18)		PrecStd	4.6	1.6	1.4	1.2	1.2	1.1	1.2	1.1	1.5	1.5	1.9	1.9	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	52.6	57.7	65.0	70.8	75.0	77.8	76.4	70.1	63.9	58.3	55.4	(19)
(20)			MCWB	51.3	49.3	50.4	55.3	60.7	63.3	65.2	66.4	63.4	59.1	56.4	53.8	(20)
(21)		2%	DB	52.0	51.1	53.9	60.2	66.3	70.1	73.3	70.8	66.6	61.6	57.0	54.3	(21)
(22)			MCWB	50.6	49.1	48.9	53.0	58.2	61.4	63.7	63.7	61.0	58.4	55.4	52.8	(22)
(23)		5%	DB	51.0	50.1	51.9	56.8	62.7	66.6	69.7	67.9	64.7	60.3	56.0	53.3	(23)
(24)			MCWB	49.7	48.5	48.2	51.0	56.3	59.6	62.4	61.9	59.8	57.7	54.4	51.8	(24)
(25)		10%	DB	50.1	49.3	50.4	54.5	59.7	64.1	67.0	65.9	63.1	59.2	55.0	52.2	(25)
(26)			MCWB	48.7	47.7	47.8	49.9	54.4	58.3	61.2	60.9	59.2	57.0	53.3	50.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	51.2	52.0	56.7	62.0	65.3	66.7	67.5	64.8	60.9	57.3	54.6	(27)
(28)			MCD B	52.4	51.5	54.2	62.8	68.5	72.4	74.4	74.1	67.8	62.5	57.6	54.9	(28)
(29)		2%	WB	51.2	50.1	50.4	53.9	59.2	62.4	64.7	64.9	62.7	59.6	56.1	53.5	(29)
(30)			MCD B	51.6	50.5	52.0	58.5	64.9	68.4	70.8	68.8	64.7	60.9	56.6	54.1	(30)
(31)		5%	WB	50.2	49.2	49.5	52.2	57.2	60.6	63.3	63.4	61.5	58.6	55.0	52.3	(31)
(32)			MCD B	50.7	49.8	50.8	55.5	61.2	65.0	67.9	66.1	63.4	59.7	55.7	53.0	(32)
(33)		10%	WB	49.2	48.2	48.6	50.8	55.5	59.1	62.1	62.2	60.4	57.6	53.7	51.1	(33)
(34)			MCD B	50.0	49.0	49.9	53.4	58.8	62.8	65.7	64.5	62.2	58.9	54.8	52.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	7.2	8.6	10.2	10.9	10.5	10.4	9.8	9.5	7.9	7.0	6.7	(35)
(36)			MCDBR	6.1	6.8	10.2	14.0	15.0	15.0	15.0	12.8	11.5	8.0	6.1	5.9	(36)
(37)			MCWBR	6.9	6.1	7.3	8.9	9.1	9.0	8.3	7.5	7.6	6.2	6.3	6.9	(37)
(38)		5% WB	MCD BR	6.2	5.8	8.2	11.6	13.0	12.9	12.9	10.7	9.3	6.7	5.8	5.9	(38)
(39)			MCWBR	7.3	6.1	6.9	8.0	8.5	8.3	7.8	7.2	7.3	6.2	6.5	7.3	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.336	0.347	0.375	0.386	0.395	0.381	0.387	0.370	0.365	0.356	0.334	0.325	(40)
(41)		taud		2.329	2.379	2.328	2.330	2.352	2.422	2.410	2.473	2.474	2.444	2.385	2.306	(41)
(42)		Ebn at Noon		208	244	258	270	272	277	273	273	261	238	209	190	(42)
(43)		Edh at Noon		21	26	32	36	37	35	35	31	28	24	19	18	(43)
(44)	All-Sky Solar Radiation	RadAvg		235	481	874	1396	1714	1864	1699	1419	1049	581	292	177	(44)
(45)		RadStd		23	43	76	98	108	157	141	94	77	47	24	22	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +0.49	(e) +1.98	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +101	(m) N/A	(46)
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page